

Hao Chen

CONTACT	Google Scholar	Homepage	Linkedin	chen4433@purdue.edu
EDUCATION	Purdue University Ph.D. in Chemical Engineering (emphasis on ML/OR)	2022 - Present Advisor: Dr. Can Li		
	University of Cambridge M.Phil. in Chemical Engineering and Biotechnology	2020 - 2021		
	University of Nottingham B.Eng. in Chemical Engineering	2016 - 2020		
APPOINTMENTS	Amazon , Seattle, WA Applied Scientist Intern, Amazon Pharmacy Science	2026 Supervisor: Dr. Xin Tang		
	• Built an agent harness that advances optimization explainability, orchestrating filesystem and shell tools, a structured context system, and an asynchronous runtime for large-scale models. • Onboarded 3 key models supporting AP’s core services and designed an evaluation pipeline. • Benchmarked the harness against a Claude Agent SDK baseline, achieving +36.5 pts success-rate gain at 1.4x lower latency and 56% fewer input tokens.			
	Amazon , Bellevue, WA Applied Scientist Intern, Last Mile Planning	2025 Supervisor: Dr. Jin Ye		
	• Developed an agentic system that interacts with optimization models at runtime and produces trustworthy natural-language explanations for business users without scientist mediation. • Delivered the first production integration on the route-allocation model behind delivery service partners handling over 90% of Amazon’s customer packages. • Now scaled to 5 models across Last Mile RaP, onboarded by the team and in daily use.			
PUBLICATIONS	[1] H Chen, C Qian, C Morris, A Lodi, C Li. Solving Max-Cut to Global Optimality via Feasibility-Preserving Graph Neural Networks. <i>arXiv</i> (2026). [2] A Ramanujam, AE Youmi, H Chen, SM Kompalli, AS Ahluwalia, S Pal, DJ Papageorgiou, C Li. SafeOR-Gym: A Benchmark Suite for Safe Reinforcement Learning Algorithms on Practical Operations Research Problems. <i>Transactions on Machine Learning Research</i> (2026). [3] H.-H Chiu, A Varghese, K Shao, Y.-C Lu, R Nahar, H Chen, Q Deng, X Bao, C Li. scChat: A large language model-powered co-pilot for contextualized single-cell RNA sequencing analysis. <i>AIChE Journal</i> (2026). [4] H Kim, H Chen, C Li, JM Lee. TalkToAgent: A multi-agent LLM Framework for natural language explanation of reinforcement learning policies. <i>Computers & Chemical Engineering</i> (2026). [5] GE Constante-Flores, H Chen, C Li. Enforcing Hard Linear Constraints in Deep Learning Models with Decision Rules. <i>Advances in Neural Information Processing Systems</i> 38 (2025). [6] H Chen, GE Constante-Flores, KSI Mantri, SM Kompalli, AS Ahluwalia, C Li. OptiChat: Bridging Optimization Models and Practitioners with Large Language Models. <i>INFORMS Journal on Data Science</i> (2025). [7] A Khan, R Nahar, H Chen, GE Constante-Flores, C Li. FaultExplainer: Leveraging large language models for interpretable fault detection and diagnosis. <i>Computers & Chemical Engineering</i> (2025). [8] H Chen, GE Constante-Flores, C Li. Physics-informed neural networks with hard linear equality constraints. <i>Computers & Chemical Engineering</i> (2024).			

- [9] **[Best Paper Award]** H Chen, GE Constante-Flores, C Li. Diagnosing infeasible optimization problems using large language models. *INFOR: Information Systems and Operational Research* (2024).
- [10] Y Yan, WI Shin, H Chen, S.-M Lee, S Manickam, S Hanson, H Zhao, E Lester, T Wu, CH Pang. A recent trend: application of graphene in catalysis. *Carbon Letters* (2021).

CONFERENCE PRESENTATIONS

Chen, H. (2026). Solving Max-Cut to Global Optimality via Feasibility-Preserving Graph Neural Networks. Paper presented at the *2026 INFORMS Annual Meeting*, San Francisco, CA

Chen, H. (2026). Neural Branch-and-Bound Solver for Global Optimization of Max Cut. Paper presented at the *2026 AIChE Annual Meeting*, Minneapolis, MN

Chen, H. (2025). Deep Learning Models with Hard Linear Inequality Constraints. Paper presented at the *2025 AIChE Annual Meeting*, Boston, MA **[CAST Director's Student Presentation Award Finalist]**

Chen, H. (2024). Self-supervised Learning for Constrained Optimization with Hard Linear Constraints. Paper presented at the *2024 INFORMS Annual Meeting*, Seattle, WA

Chen, H. (2024). GPU Accelerated Approximation Algorithm for Multi-Parametric Linear Programming. Paper presented at the *2024 AIChE Annual Meeting*, San Diego, CA

Chen, H. (2024). Physics-Informed Neural Networks with Hard Linear Equality Constraints. Paper presented at the *2024 AIChE Annual Meeting*, San Diego, CA

Chen, H. (2024). Diagnosing Infeasible Optimization Problems Using Large Language Models. Paper presented at the *2024 AIChE Annual Meeting*, San Diego, CA

HONORS AND AWARDS

People's Choice Award, Global AI Solutions Expo, Amazon	2026
INFOR Best Paper Award, 67th CORS Annual Conference	2026
Bilsland Dissertation Fellowship, Purdue University	2026
Purdue Graduate Student Government (PGSG) Travel Grant	2025
Ryland Travel Grant, School of Chemical Engineering, Purdue University	2025
Purdue Graduate Student Government (PGSG) Travel Grant	2024
Chao Travel Grant, School of Chemical Engineering, Purdue University	2024
Nottingham Engineering Excellence Scheme (Top 1.0%), University of Nottingham	2019
British Petroleum Prize, University of Nottingham	2019
Provost's Scholarship (Top 1.5%), University of Nottingham	2018
Dean's Scholarship (Top 10%), University of Nottingham	2017

PROFESSIONAL ACTIVITIES

Session Chair: *INFORMS Annual Meeting* (2026)

Reviewer: *NeurIPS* (2026); *Manufacturing & Service Operations Management* (2026); *Computers & Chemical Engineering* (2025-2026); *Digital Chemical Engineering* (2025-2026); *Applied Energy* (2025)

SKILLS

Languages: Python, Julia, SQL

ML/LLM Stack: PyTorch, PyG, SB3, HuggingFace, Google ADK, Claude Agent SDK

OR Stack: JuMP, Gurobi, Xpress, Pyomo, CPLEX

Others: AWS EC2, ECS, Docker, Lambda, DynamoDB, S3